

Summary of Revisions: From Initial Draft to Final Version

This document outlines the key updates, corrections, and theoretical additions made to transition from the initial draft to the final version of our NMSA report. The revisions heavily incorporate your recent review notes to elevate the mathematical and physical rigor of the text.

Homework 1: FEM Discretization & Acoustics

- **Dispersion Curve Correction (Section 1.4):** In the original draft, the k_h/k curve bent downwards (underestimating the wavenumber). Following your notes and the Taylor series expansion for the consistent-mass matrix, the MATLAB code was fixed. The final curve correctly bends **upwards** ($k_h/k > 1$). We also added a high-level remark explaining the curve's behavior near the Nyquist limit ($h/\lambda \rightarrow 0.5$), where it departs from the parabola and lands at ≈ 1.10 .
- **PML Visualization (Section 1.5):** The original plots stopped at the physical boundary ($x=1$). We extended the plotting domain to $x=1.2$. This visually proves the **exponential amplitude decay** inside the absorbing layer, while showing that the phase continues to decrease linearly (confirming it remains an outgoing traveling wave while dissipating energy).

Homework 2: Linearized Shallow Water Equations (SEM)

- **Reflecting Walls Assembly (Section 2.5):** As per your comment, we retained and defended the explicit explanation of how periodic conditions are replaced by reflecting walls at the matrix level (removing the periodic link and zeroing out the q degrees of freedom).

Homework 3: Nonlinear SWE (FVM)

- **Structural Reorganization:** The chapter was restructured into exactly **three main sections** to perfectly mirror the assignment requirements, moving all the heavy mathematical FVM theory into Section 3.1 to make the results (Section 3.2) and the modified flux (Section 3.3) easier to read.
- **Adaptive CFL Condition:** We explicitly added the adaptive time-step formula $\Delta t^n = \nu \frac{\Delta x}{\max_i S_i^n}$ to show we understand how explicit FVM time-marching relies on local wave speeds.
- **Quantitative Consistency Check:** We implemented your excellent analytical Rankine-Hugoniot estimates for the shock speed in the dam break problem (e.g., estimating $\sigma \approx 3.1$ m/s). This mathematically proves *why* the shock hits the right wall before $T=1$ s, rather than just observing it from the plot.
- **The "Ill-Posedness" Discovery (Section 3.3):** We highlighted your brilliant critical remark regarding the modified logarithmic pressure flux. We mathematically demonstrated that for $h < h_{\text{ref}} e^{-1}$, the eigenvalues become complex, making the system **ill-posed** in the right half of the Dam Break domain. We clarified that the MATLAB code doesn't crash solely due to the artificial numerical viscosity of the Rusanov solver.
- **Comparison of Celerities:** We added the formal inequality showing $c_{\text{mod}} < c_{\text{std}}$ to analytically justify why the modified shock travels significantly slower than the standard hydrostatic one.